Govt. Degree College, Rajgir

Bachelor Of Computer Application (BCA)

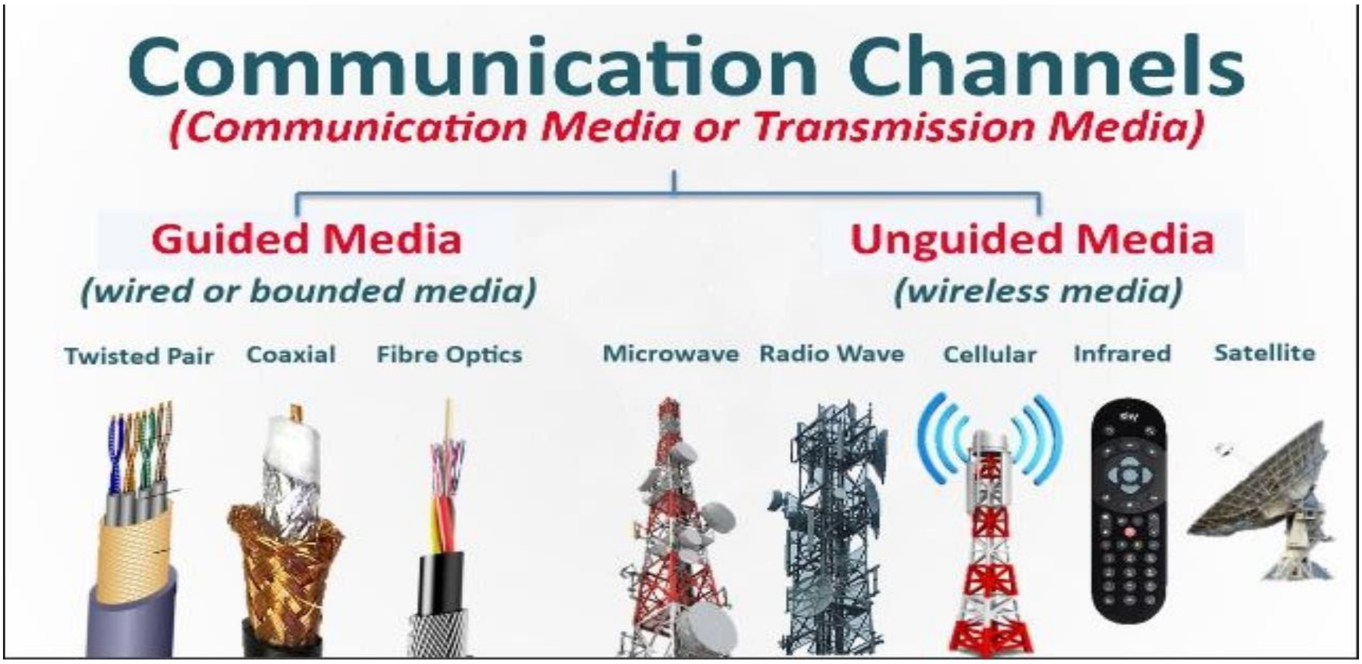
Part -2 (Paper-3) **– Hira Kumar**

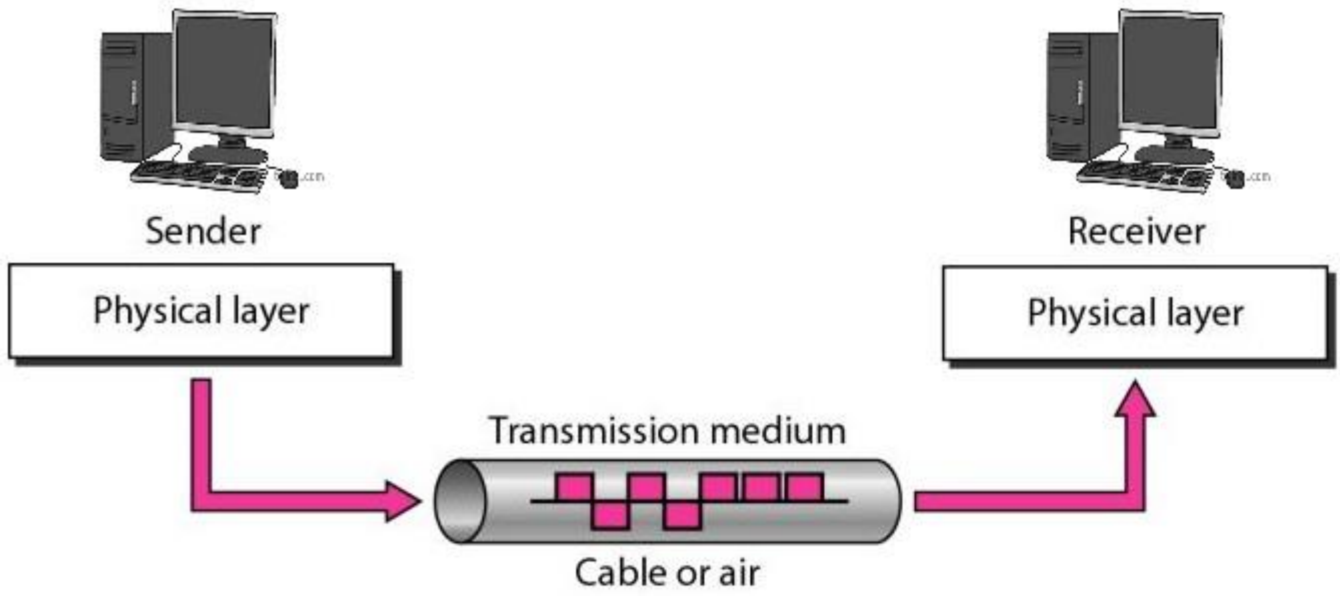
Computer Networking

Data Communication

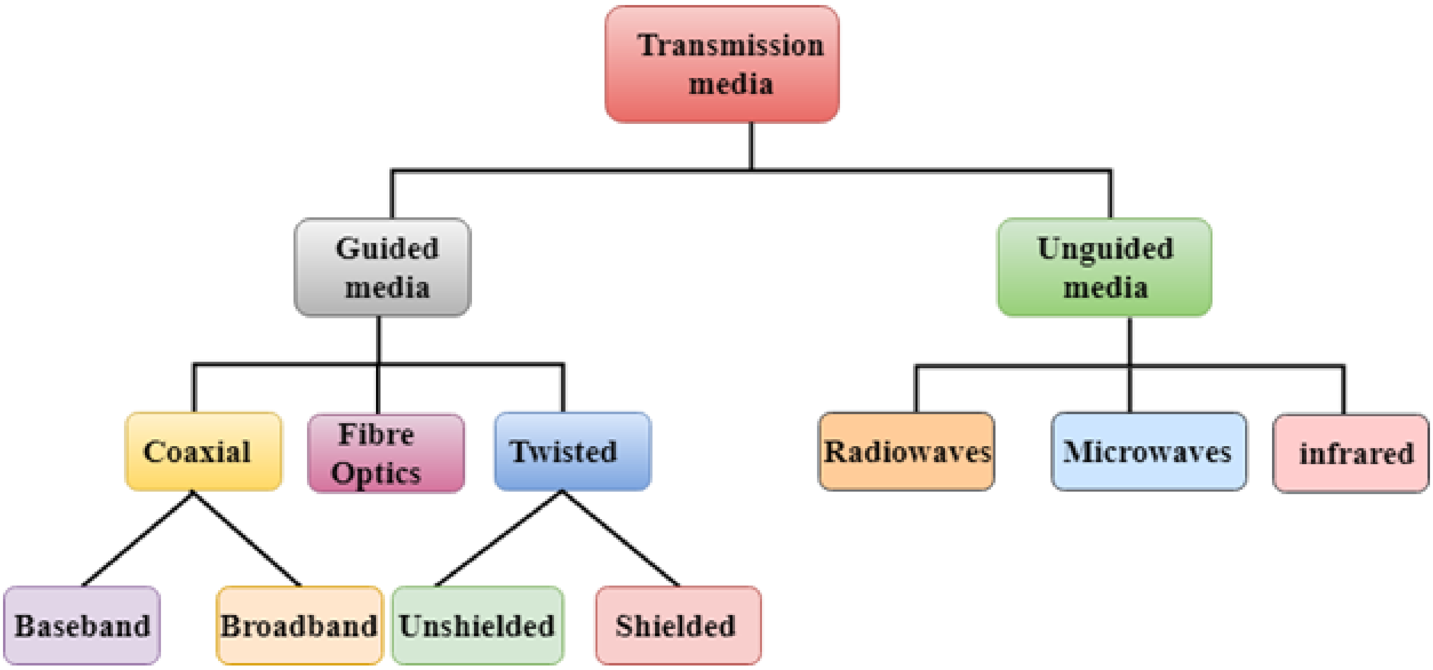
❖ Communication Media / Transmission Media / Communication Line

- A part of physical layer in networking / Internet.
- The transmission medium is the means by which data/signal is transmitted/carried from one location to another location.
- A transmission medium can be defined as anything that can carry digital information from a source to a destination.
- Any message can be sent as data that can be translated into binary digits(0,1).The signal is encoded/encrypted with these digits.So(ताकि), data is sent to a network, the network divides into packets, and the packets are formed by the network.
- Just like A mail carrier truck or train/aeroplane could be the delivery medium for a written letter/goods. Similarly,data transfer one nodes to another nodes.





Types of Transmission Media



1. Guided Media / Bounded Media

- In guided media, data signals flow through wires.
- Data is transmitted through these wires to a particular patch(पथ/रास्ता).
- These wires are made of copper,tin or silver.
- Guided media also known as Bounded media, which are those that provide a conduit (नलिका) from one device to another device.

➢ **Bounded transmission media / Guided Media are three types**

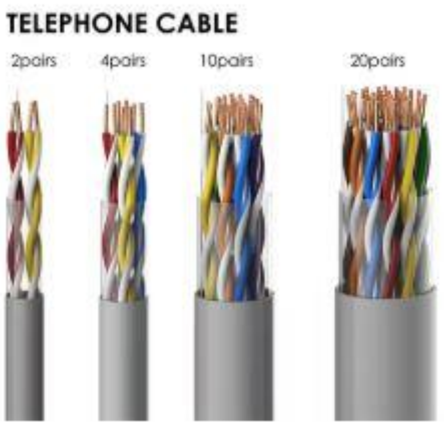
A. Twisted-pair cable

B. Coaxial cable

C. Fiber-optic cable

A. Twisted Pair Cable/wire

- The wires is twisted-twisted (मुड़-मुड़) together in pairs(जोड़ा).
- The less expensive and most widely used guided transmission medium.
- Twisted pair (TP) maybe used to transmit both analog and digital signal.
- It is lightweight, cheap, can be installed easily, and they support many different types of network.
- Over longer distances, cables may contain hundreds of pairs.
- For analog signals amplifiers are required about every 5 to 6 kms. For digital signals repeaters are required every 2 to 3kms. (Analog – amplifiers , Digital – repeaters).



Application/Use

✓ It is the most commonly used medium of telephone.

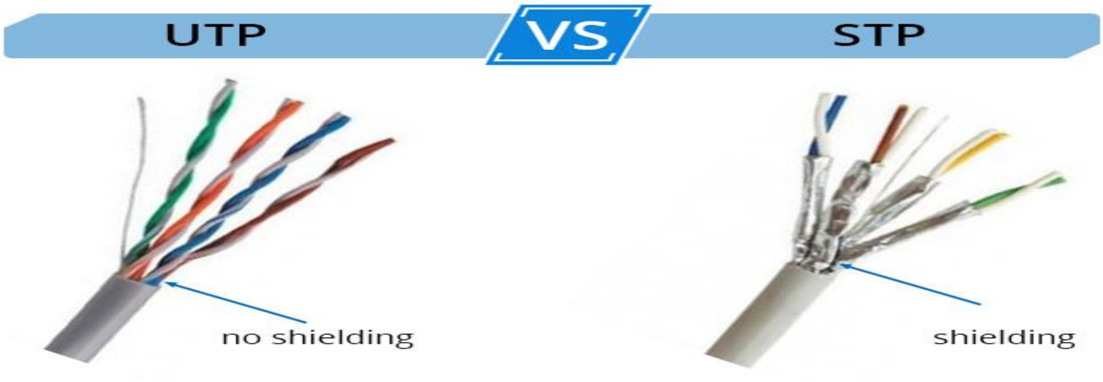
✓ In the telephone system individual residential telephone sets are connected to the local telephone exchange or to end office by twisted pair wire.

✓ It is also commonly used within a building for LAN(Local area network) / Ethernet.

➢Two types of Twisted pair cable

i) Unshielded twisted pair (UTP)

ii) Shielded twisted pair (STP)

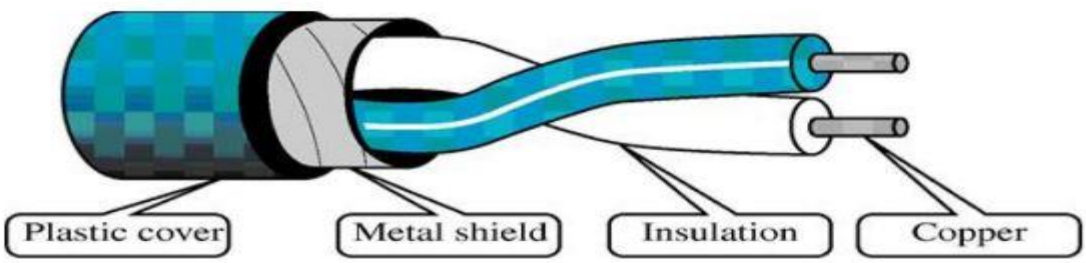


i) UTP (Unshielded twisted pair)

- Usually consists of two copper wires wrapped in individual plastic insulation.
- UTP cables are the most common telecommunications medium.
- The frequency range of the twisted pair cables enable both voice and data transmission.
- UTP cables consist of 2 or 4 pairs of twisted cable. Cable with 2 pair use RJ11 (Telephone) connector and 4 pair cable useRJ-45(PC/Laptop/router,…) connector.

ii) STP (Shielded twisted pair)

- The only difference between STP and UTP is that STP cables have a shielding in usually of aluminum or polyester material between the outer jacket and wire.
- The metal mesh around the insulated wires eliminates crosstalk. Crosstalk occurs when one line picks up some of the other signals traveling down another line.



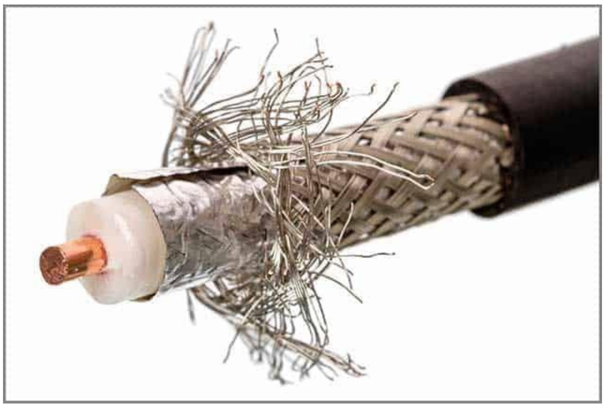
HIRA KUMAR

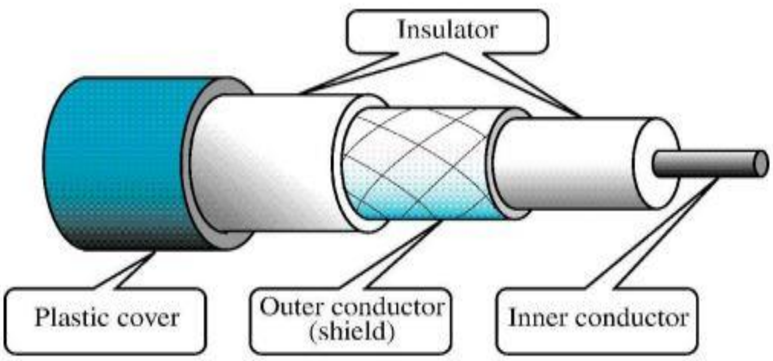
➢ Difference B/W UTP & STP

Parameters	UTP	STP
Full Form	Unshielded Twisted Pair	Shielded Twisted Pair
Structure	cable with wires that are twisted together.	Twisted pair cable enclosed in foil / shield.
Cost	Cheaper than STP	Costlier than UTP
Weight	Lighter than STP	Heavier than UTP
Noise & interference	Prone to Noise and interference nature	Less prone to noise and interference
Data Speed	Supports slower speed than on STP	Support higher speed than UTP
Grounding of cable	Not required	Required
Target deployments	Locations less prone to interference like offices and homes.	Locations prone to interference like factories and airports

B. Co-axial cable

- The name coaxial is because it contains two conductors that are parallel to each other and share common axis.
- Inner conductor is made of copper which is surrounded by PVC insulation.
- The outer conductor is metal foil(पत्ति), mesh(जाली) or both.
- Outer metallic conductor is used as a shield against noise.
- The outermost part is the plastic cover which protects the whole cable.
- Co-axial cable is much less susceptible(अतिसंवेदनशील) to interference and cross talk than the twisted pair.
- Co-axial cable is used to transmit both analog and digital signal.





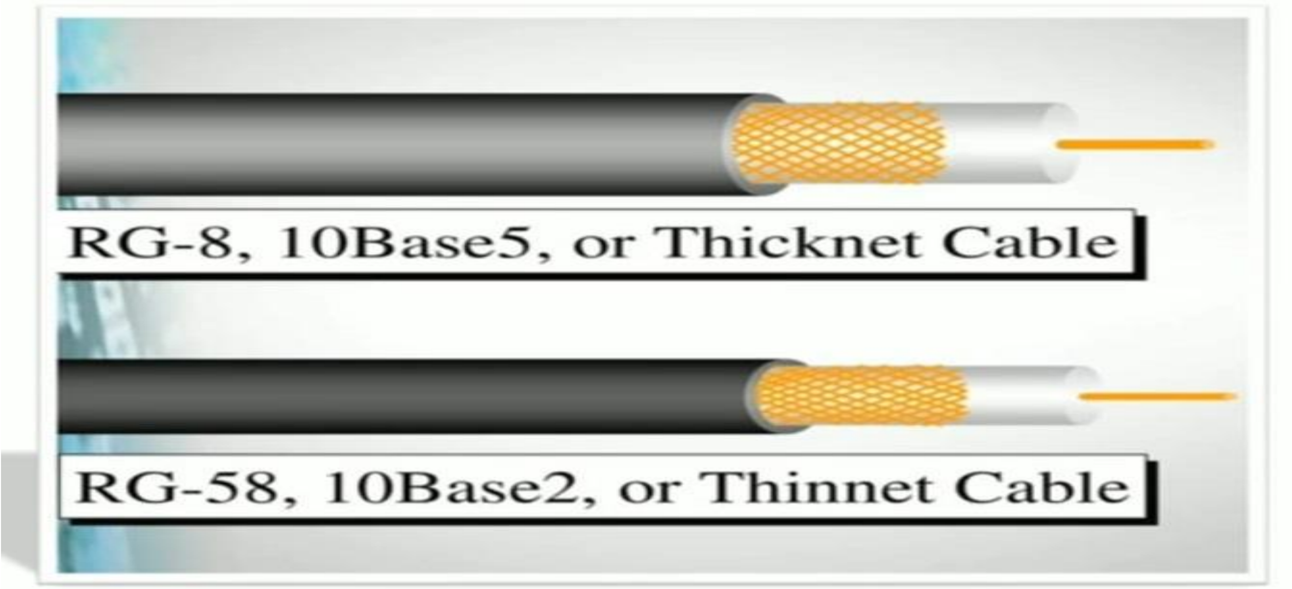
➢ Application /Use

- ✓ TV distribution cable /Dish Cable
- ✓ Long distance telephone transmission
- ✓ Short distance communication links (pc to pc)
- ✓ LAN

➢ Co-axial cables usually for data transmission are of two types.

- 10 base 2 thin net - Node capacity per segment for 10 base 2. (speed 10 mbp/s, range 200 meter)
- 10 base 5 thick net- Node capacity for 10 base 5.(speed 10 mbp/s, range 500 meter)

  - ✓ Analog – amplifiers (a speed -2xa/3xa speed) , Digital – repeaters (x speed – x speed)



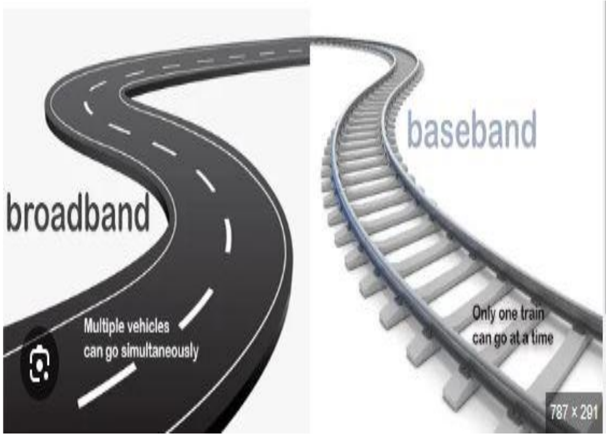
➢ BaseBand

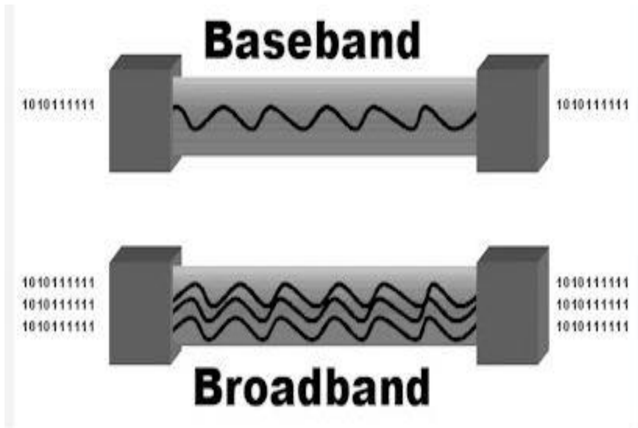
- It is mostly used for LAN (local area network).
- Baseband transmits a single signal at a time with very high speed.
- The major drawback is that it needs amplification after every 1000 feet / 300 meter.

➢ BroadBand

- This uses analog transmission on standard cable television cabling.
- It transmits several simultaneous signal using different frequencies.

- It covers large area when compared with Baseband Coaxial Cable





➢ Bandwith

- The maximum amount of data transmitted over an internet connection in a given amount of time.
- OR we can say that bandwith how many data transfer at a one time and to what distance.
- Time measure in MBPS(megabite per second).

➢ Throughput

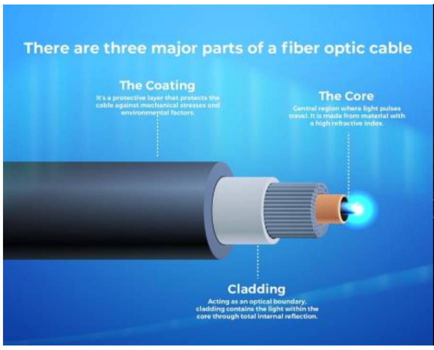
- Throughput indicates how much data was actually transferred one end to another end. (e.g - send 30mbps but recive 25 mbps)

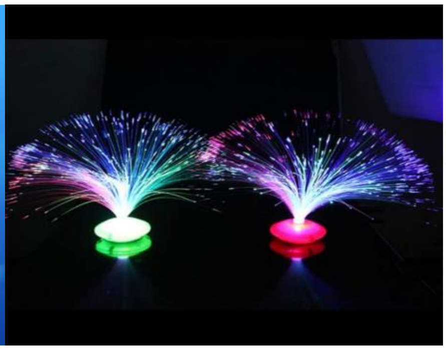
➢ Latency

- Latency means how many time taken when data transmit source to reciver.i.e. depended how many traffic on media/wire.
- Latency measure in millisecond(ms).

C. Fiber-optic cable OR Optical Fiber

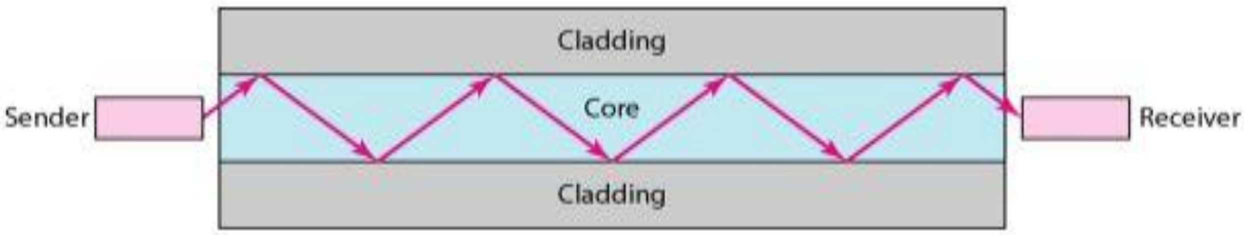
- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.
- A light pulse/vibration can be used to signal a one (1) bit.The absence of a pulse signals a zero(0).
- The bandwidth of an optical transmission system is very high.
- The transmission medium is an ultra-thin fiber of glass






An optical fiber has an cylindrical shape and consists of 3 concentric section

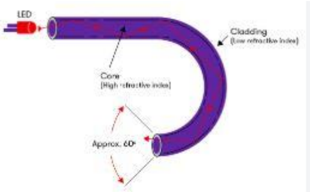
(i) Core **(ii) Cladding** **(iii) Jacket**



1. **Core:-** It's the inner most section is made of glass or plastic and is surrounded by its own cladding. The core diameter is in the range of 8 to 50 µm.
2. **Cladding: -** A glass or plastic coating that has optical properties different from those of the core having a diameter of 125 µm. The cladding acts as a reflector to light that would otherwise escape the core.
3. **Jacket: -** The outer most layers surrounding caddied fiber is the jacket. Jacket is composed of plastic or other material layer to protect against moisture, cut, crushing and other environmental dangers.

➢ Optical Fiber Communication

- A transmitter (Light Source) at senders end sends a Light across the fiber.
- A receiver at the other end makes use of Light Sensitive transistor to detect the absence or presence of light to indicate 0 or 1..



➢ Two different types of light sources are used in fiber optic system.

- The Light Emitting Diode (LED)
- Injection Laser Diode (ILD).

➢ Application/Use

- ✓ Telephones
- ✓ including cellular wireless Internet
- ✓ LANs - local area networks, WAN
- ✓ CCTV Camera - for video,
- ✓ Medical Application
- ✓ Transportation – smart lights and highways
- ✓ Military

➢ Difference / Characteristics of cable media:-

Factor	UTP	STP	Co-axial	Fiber optics
Cost	Low	Moderate	Moderate	Highest
Installation	Easy	Fairly easy	Fairly easy	Difficult
Data rate	1 to 155 mbps	1to 155 mbps	500 mbps	2 GBPS
Node capacity	2	2	30-100	2
Attenuation	High(100's of meter)	High(100's of meter)	Lower (range of few km's)	Lowest (10 Km's)
EMI	Most vulnerable	Less vulnerable than UTP	Less vulnerable than UTP	Not effected by EMI
Bandwidth	Low	Moderate	Moderatly high	Very high
Signals	Electrical	Electrical	Electrical	Light

=====================HIRA KUMAR========================

HIRA KUMAR